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Step 2.2: For all the digits write powers of 2 from left to right starting from $2^{-1}, 2^{-2}, 2^{-3}$..... 2^{-n},

now write the equivalent weight.

Step 2.3: Multiply each digit with its corresponding weight

Step 2.4: Add all the values which you obtained in Step 2.3

Table 2.5 Positional notation and weight

Positional notation	Weight
2^{-1} (1/2)	0.5
2^{-2} (1/4)	0.25
2^{-3} (1/8)	0.125
2^{-4} (1/16)	0.0625
2^{-5} (1/32)	0.03125
2^{-6} (1/64)	0.015625
2^{-7} (1/128)	0.0078125

Step 3: To get final answer write the integral part (after conversion), followed by a decimal point(.) and the answer arrived at Step 2.4

Example

Convert the given Binary number $(11.011)_2$ into its decimal equivalent Integer part $(11)_2 = 3$ (Refer **table 2.2**)

2^1	2^0		2^{-1}	2^{-2}	2^{-3}
↑	↑		↑	↑	↑
1	1	.	0	1	1

$3 + . (0\times0.5 + 1\times0.25 + 1\times0.125)$
$= 3.375$
$(11.011)_2 = (3.375)_{10}$

Workshop

4. Convert the given Binary number into its equivalent Decimal, Octal and Hexadecimal number.

1) 101110101
2) 1011010
3) 101011111

2.4.9. Octal to Decimal Conversion

To convert Octal to Decimal, we can use positional notation method.

1. Write down the Octal digits and list the powers of 8 from right to left(Positional Notation)
2. For each positional notation of the digit write the equivalent weight.
3. Multiply each digit with its corresponding weight
4. Add all the values

Example

Convert $(1265)_8$ to equivalent Decimal number

Weight	512	64	8	1
Positional Notation	8^3	8^2	8^1	8^0
Given number	1	2	6	5

$(1265)_8 = 512\times1 + 64\times2 + 8\times6 + 1\times5$
$= 512 + 128 + 48 + 5$
$(1265)_8 = (693)_{10}$